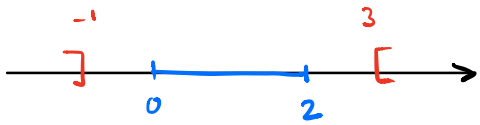
Questions de Cours

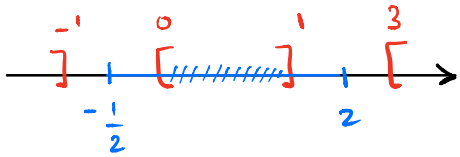
$X =]-1,3[\quad \alpha = 0 \quad \beta = 2$

X est connexe par arc



ok !

$X =]-1,0[\cup]1,2[$



X

• Prop 2.1 $\quad A \neq \emptyset \; ; \; A \subset \mathbb{R} \; , \; \delta : A \longrightarrow \mathbb{R}_{>0}$

$$[a,b] = \bigcup_{x \in A}]x - \delta(x), x + \delta(x)[$$

Questions sur les exercices

Exercice 1

$$\frac{\sqrt[5]{93}\,\sqrt[15]{3}\,\sqrt[3]{9}\,\sqrt{12}}{(\sqrt[3]{\sqrt{3}})^2\,\sqrt{18}\,\sqrt[5]{27}\,\sqrt{6}} = \frac{\sqrt[5]{31}}{3}$$

$$\frac{\sqrt[5]{93}\,\sqrt[15]{3}\,\sqrt[3]{9}\,\cancel{2}\,\cancel{\sqrt{3}}}{\sqrt[3]{3}\,\underline{\underline{3}}\,\cancel{\sqrt{2}}\,\sqrt[5]{27}\,\cancel{\sqrt{3}}\,\cancel{\sqrt{2}}} = \frac{\sqrt[5]{93}\,\sqrt[15]{3}}{\sqrt[3]{3}\,\sqrt[5]{27}\cdot 3} = \sqrt[5]{\frac{93}{27}}\,\frac{\sqrt[15]{3}}{\sqrt[3]{3}}\,\sqrt[3]{9}\,\frac{1}{3}$$

$$= \sqrt[5]{\frac{\cancel{3}\times 31}{\cancel{3}\times 9}}\,\sqrt[3]{\frac{9}{3}}\,\frac{\sqrt[15]{3}}{3}$$

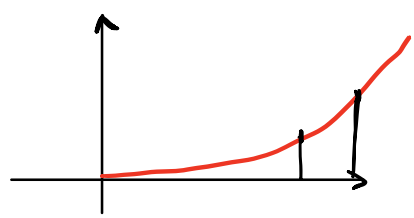
$$= \sqrt[5]{\frac{31}{9}}\,\underbrace{\sqrt[3]{3}\,\sqrt[15]{3}}\,\frac{1}{3}$$

$$\underbrace{3^{1/3}\,3^{1/15}}_{\substack{\shortparallel \\ \underbrace{3^{\frac{1}{3}+\frac{1}{15}}}_{3^{2/5}}}}$$

$$= \sqrt[5]{\frac{31}{9}}\; 3^{2/5}\,\frac{1}{3} = \left(\frac{31}{9}\right)^{1/5}\underbrace{(3)^{2/5}}_{\sqrt[5]{3^2}}\,\frac{1}{3} = \left(\frac{31\times\cancel{9}}{\cancel{9}}\right)^{1/5}\frac{1}{3}$$

<u>Exercice 2</u>

$$\underset{\substack{\shortparallel \\ 2\underbrace{\sqrt{2}}_{\simeq 1{,}41}}}{(\sqrt{2})^3} \qquad (\sqrt[3]{3})^3 = 3$$



Comme X^3 est croissante on en déduit que $\sqrt[3]{3} > \sqrt{2}$

$$(\sqrt{2})^5 = (\sqrt{2})^4\sqrt{2} = \underbrace{4\underbrace{\sqrt{2}}_{\simeq 1{,}41}}_{\simeq 5{,}6\cdots} \qquad (\sqrt[5]{5})^5 = 5$$

Comme X^5 est croissante sur $\mathbb{R}^+$ on a $\sqrt{2} > \sqrt[5]{5}$.

<u>Exercice 4</u>

$$\sqrt[3]{20+14\sqrt{2}} = n+\sqrt{2}$$

$$20+14\sqrt{2} = (n+\sqrt{2})^3 = n^3 + 3\sqrt{2}\,n^2 + 6n + 2\sqrt{2}$$

$$n^3 + 3\sqrt{2}\,n^2 + 6n - 12\sqrt{2} - 20 = 0 \qquad \underline{\underline{n = 2}}$$

Exercice 5

$$(x+1)^2 - |x-2| \geq 0$$

$$(x+1)^2 - \sqrt{(x-2)^2} \geq 0$$

$$(x+1)^2 \geq \sqrt{(x-2)^2}$$

$$(x+1)^4 \geq (x-2)^2$$

alternative

$$-|x-2| \geq -(x+1)^2$$

$$|x-2| \leq (x+1)^2$$

$$-(x+1)^2 \leq x-2 \leq (x+1)^2$$

$a^2 - b^2 = (a-b)(a+b)$

$$\left((x+1)^2\right)^2 - (x-2)^2 \geq 0$$

$$\left((x+1)^2 - (x-2)\right)\left((x+1)^2 + (x-2)\right) \geq 0$$

$$(x^2+1+2x-x+2)(x^2+1+2x+x-2) \geq 0$$

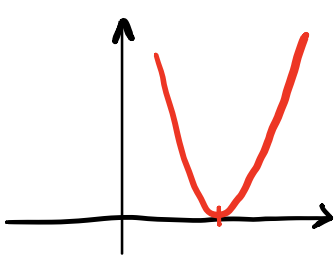
$$(x^2+3+x)(x^2-1+3x) \geq 0$$

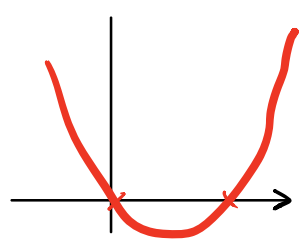
Exercice 6

$$P(x) = x^2 + 2(m-1)x + (m^2+7)$$

$\{m \in \mathbb{R} \text{ tq } P(x) \geq 0\}$

$\underbrace{\{m \in \mathbb{R},\ P(x) < 0\}}_{= \emptyset}$





$\Delta_{P(x)} \leq 0$

$\Delta_{P(x)} = (2(m-1))^2 - 4(m^2+7)$

$\Delta_{P(x)}$ est un polynôme du 2nd degré en M.

$$\Delta = 4(M-1)^2 - 4M^2 - 28 = 4(\cancel{M^2}+1-2M) - \cancel{4M^2} - 28$$

$$= -24 - 8M$$

$$-24 - 8M \leq 0$$

$$-3 \leq M \qquad \{M \in \mathbb{R} \mid \forall x \in \mathbb{R}, P(x) \geq 0\} = \{M \geq -3\}$$

<u>Exo 7</u> Distinguer $5m - 6 \geq 0$

$$5m - 6 < 0$$

<u>Exercice 8</u> 4) $\forall n \in \mathbb{N}, 0 \leq \frac{n}{n+1} \leq 1$

$$1 \leq E\left(\frac{n+4}{n}\right) \leq 5 \qquad 1 \leq \frac{n+4}{n} = 1 + \frac{4}{n} \leq 5$$

$$\underset{n=0}{0} \leq \frac{n}{n+1} E\left(\frac{n+4}{n}\right) \leq \underset{n=1}{5/2}$$ "produit des deux plus grand"

$5, 3, 1 + \frac{1}{3} \simeq 1{,}33, 1{,}25$

<u>Exercice 9</u> $\sup(a,b) = \begin{cases} a & \text{si } a \geq b \\ b & \text{si } b > a \end{cases}$